Kasra Sinaei

Robotic Software Developer | **Control & Physical AI** Researcher | **Authorized** to work in the **US**
kassra-sinaei.github.io | kasra.sinaei@gmail.com | (582)203-9717 | kasra@psu.edu

EDUCATION

PENNSYLVANIA STATE UNIVERSITY
PhD in Electrical Engineering
Thesis: Safety Guarantee for Learning Policies via Barrier Functions
May 2027 | State College, PA

PENNSYLVANIA STATE UNIVERSITY
MSc. in Electrical Engineering
Paper: Model-free Safety-critical Position Control of Robots
May 2024 | State College, PA

UNIVERSITY OF TEHRAN
BSc. in Mechanical Engineering
Capstone Project: Motion Planning & Control of Wheeled-legged Robot
Jan 2022 | Tehran, Iran
Cum. GPA: 17 / 20

SKILLS

PROGRAMMING
C++ • C • Python • MATLAB • Java • C# • Shell

SOFTWARE DEVELOPMENT
ROS1/2 • Git • Docker • CUDA • .NET • AWS • Kubernate • Web Stack • Android Studio

ROBOTICS & SIMULATION
Nvidia IsaacSim • MuJoCo • Gazebo • PyBullet • Pinochio • Moveit • ROS Nav

COMPUTER VISION & AI
PyTorch • OpenCV • CUDA • Hugging-face • Transformers • Lerobot • YOLO • W&B

THEORETICAL
Nonlinear Control • Convex Optimization • Adaptive/Robust Control • VLM/ VLA • RL • SLAM • Motion Planning • MPC • Trajectory Optimization

CAD/CAE
SolidWorks • ANSYS • CATIA • AutoCAD

TEACHING ASSISTANT

Intro. to Linear Control Systems (EE380)	SP-2026
Continuous-Time Linear Systems (EE350)	FA-2025
Intro. to Digital Control Systems (EE482)	SP-2024
AI and ML in Mechanical Engineering	FA-2021
Electrical Circuits and Electrical Machines	FA-2020

LANGUAGE SKILLS

ENGLISH
Full Professional Proficiency C2 (CEFR)

SPANISH
Elementary Proficiency A2 (CEFR)

INDUSRTY EXPERIENCE

CHEWY CO. | AI Research Scientist Intern
Jun 2026 - Sep 2026 | Boston, MA
Developed **Semantic Safety Filters** for mobile manipulators powered by physical AI policies. Leverged **3D Scene Graphs** for retrieving semantic information from the environment. Deployed **VLA** for long horizon warehouse tasks.

ALPHAZ INC. | Robotics Software Engineer Intern
May 2025 - Aug 2025 | Pasadena, CA
Deployed **trajectory planning** algorithm for dual-arm **mobile manipulator** in real-time. Built high-reliability ROS2 **perception pipelines** for state estimation and object detection. Integrated research code into production navigation stack.

RESEARCH EXPERIENCE

CARL | Graduate Researcher
Aug 2022 - Now | State College, PA
developing **theoretical guarantees** for safety-critical control systems and implementing model-free **safety** filters by bridging the gap between classical **control theories** and modern **foundation model policies**. Validated the proposed methods in simulation and hardware tests.

CAST | Undergraduate Researcher
Sep 2019 - Mar 2022 | Tehran, Iran
Developed core software for full-size **humanoid robot** at CAST. Implemented **low-level drivers**, **state estimation** fusing IMU/encoders, and closed-loop balance controllers. Designed gait controllers for traversing uneven terrain.

AWARDS & HONORS

MELVIN P. BLOOM MEMORIAL GRADUATE FELLOWSHIP | $850 | 2024
Awarded to outstanding graduate students in Electrical Engineering based on academic excellence and research potential.

MILTON & ALBERTHA LANGDON MEMORIAL GRADUATE FELLOWSHIP | $810 | 2023
Merit-based fellowship recognizing exceptional graduate student performance and research contributions.

DR. ARTHUR H. WAYNICK GRADUATE SCHOLARSHIP IN EE | $3,147 | 2022
Prestigious scholarship awarded to top-performing graduate students in Electrical Engineering department.

COLLEGE OF ENGINEERING GRADUATE FELLOWSHIP | $3,000 | 2022-2023
Competitive fellowship supporting outstanding graduate students pursuing advanced degrees in engineering disciplines.

GOVERNMENTAL BSC. SCHOLARSHIP | Full Tuition Waiver | 2017-2021
National merit-based scholarship covering complete undergraduate tuition for academically distinguished students.

PUBLICATIONS

[1] **K. Sinaei** and D. Ebeigbe, **Control barrier functions for adaptive stabilization of fully unknown underactuated nonlinear systems**, in *Conference on Decision and Control (CDC)*, IEEE (Under Review), 2027.

[2] H.-C. Wu, X. Zhang, **K. Sinaei**, R. Abnavi, K. Weerakoon, C. Bradley, S. Fakoorian, J. Li, and D. Ebeigbe, **Strategic and reactive stream partitioning for path-efficient lidar-based obstacle avoidance**, in *International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2026.

[3] **K. Sinaei**, H.-C. Wu, and D. Ebeigbe, **Safe vision language action model via barrier enhanced flow matching**, *IEEE Robotics and Automation Letters (Under Review)*, 2026.

[4] **K. Sinaei** and D. Ebeigbe, **Safe adaptive control with vanishing conservativeness for robotic systems with unknown dynamics via barrier functions**, *IEEE Robotics and Automation Letters*, 2026.

[5] **K. Sinaei**, K. Weerakoon, C. Bradley, S. A. Fakoorian, and D. Ebeigbe, **Motion planning for mobile manipulators navigating doorways via model predictive path integral**, in *Modeling Estimation and Control Conference (MECC)*, American Society of Mechanical Engineers, 2026.

[6] **K. Sinaei** and D. Ebeigbe, **Safe robust control of nonlinear systems with uncertain regressor and parameter vector**, in *2025 IEEE Conference on Control Technology and Applications (CCTA)*, 2025.

[7] **K. Sinaei**, H.-C. Wu, and D. Ebeigbe, **Safety-critical position control of robots: A model-free approach**, in *2025 American Control Conference (ACC)*, 2025.

[8] P. Abdolahnezhad, A. Yousefi-Koma, A. Vedadi, **K. Sinaei**, B. Maleki, and M. Shafiee, **Online bipedal locomotion adaptation for stepping on obstacles using a novel foot sensor**, in *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*, IEEE, 2022.

[9] A. Vedadi, **K. Sinaei**, P. Abdolahnezhad, S. S. Aboumasoudi, and A. Yousefi-Koma, **Bipedal locomotion optimization by exploitation of the full dynamics in dcm trajectory planning**, in *2021 9th RSI International Conference on Robotics and Mechatronics (ICRoM)*, IEEE, 2021.

[10] **K. Sinaei** and M. Yazdi, **Pid controller tuning with deep reinforcement learning policy gradient methods**, in *Proceedings of the 29th International Conference of Iranian Society of Mechanical Engineers, Tehran, Iran*, 2021.

SELECTED PROJECTS

More details on open source projects available at **kassra-sinaei.github.io**.

SAFE VLA VIA BARRIER ENHANCED FLOW MATCHING | Thesis Research

2026 | CARL, Penn State University

Developing **vision-language-action (VLA)** models for robotic manipulation with safety guarantees. Modified the base Flow Matching algorithm to effectively bridge **foundation models** and CBF to ensure safe deployment of learned policies. Validated through hardware experiments on manipulation tasks.

AUTONOMOUS DOOR OPENING WITH MOBILE MANIPULATORS | Internship Project

2025 | AlphaZ Inc.

Developed a fully autonomous system for door handle **detection** and **manipulation** using mobile manipulator. Implemented a novel motion planning algorithm, **real-time** control and **perception** pipelines.

MODEL-FREE SAFETY-CRITICAL ROBOT CONTROL | Master's Thesis Project

2022 - 2024 | Carl, Penn State University

Created a model-free safety control framework using **Control Barrier Functions** for position controlled robotic systems. Validated safety guarantees through **Lyapunov** analysis and **Set invariance** principles.

HUMANOID ROBOT GAIT CONTROLLER | Undergraduate Capstone Project

Jan 2020 - May 2022 | University of Tehran

Developed **state estimator** and closed-loop **balance controllers** for full-size **humanoid**. Implemented adaptive gait patterns for traversing uneven terrain under uncertainty.

DEEP REINFORCEMENT LEARNING FOR PID TUNING | Independent Research

2021 | University of Tehran

Implemented **RL** algorithms (PPO, A3C) for automatic PID controller gain optimization. Built **training framework** with custom gym environments and reward engineering pipelines.

CETIFIED TRAININGS

- NVIDIA DLI: Fundamentals of Deep Learning
- AWS Certified Cloud Practitioner
- ROS Industrial Training
- Advanced Programming with CPP